

JUnit/Mockito Exercise 7: SLF4J Logging Logback

Logging using SLF4J

Exercise 1: Logging Error Messages and Warning Levels

Task: Write a Java application that demonstrates logging error messages and warning levels using SLF4J.

Step-by-Step Solution:

1. Add SLF4J and Logback dependencies to your `pom.xml` file:

```
<dependency>
  <groupId>org.slf4j</groupId>
  <artifactId>slf4j-api</artifactId>
  <version>1.7.30</version>
</dependency>
<dependency>
  <groupId>ch.qos.logback</groupId>
  <artifactId>logback-classic</artifactId>
  <version>1.2.3</version>
</dependency>
```

2. Create a Java class that uses SLF4J for logging:

```
import org.slf4j.Logger;
import org.slf4j.LoggerFactory;

public class LoggingExample {
    private static final Logger logger = LoggerFactory.getLogger(LoggingExample.class);

    public static void main(String[] args) {
        logger.error("This is an error message");
        logger.warn("This is a warning message");
    }
}
```

Exercise 2: Parameterized Logging

Task: Write a Java application that demonstrates parameterized logging using SLF4J.

Step-by-Step Solution:

1. Add SLF4J and Logback dependencies to your `pom.xml` file:

```
<dependency>
  <groupId>org.slf4j</groupId>
  <artifactId>slf4j-api</artifactId>
  <version>1.7.30</version>
```

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```
</dependency>
```

```
<dependency>
```

```
  <groupId>ch.qos.logback</groupId>
```

```
  <artifactId>logback-classic</artifactId>
```

```
  <version>1.2.3</version>
```

```
</dependency>
```

2. Create a Java class that uses SLF4J for parameterized logging:

Write code for this.

Exercise 3: Using Different Appenders

Task: Write a Java application that demonstrates using different appenders with SLF4J.

Step-by-Step Solution:

1. Add SLF4J and Logback dependencies to your `pom.xml` file:

```
<dependency>
```

```
  <groupId>org.slf4j</groupId>
```

```
  <artifactId>slf4j-api</artifactId>
```

```
  <version>1.7.30</version>
```

```
</dependency>
```

```
<dependency>
```

```
  <groupId>ch.qos.logback</groupId>
```

```
  <artifactId>logback-classic</artifactId>
```

```
  <version>1.2.3</version>
```

```
</dependency>
```

2. Create a `logback.xml` configuration file to define different appenders:

```
<configuration>
```

```
  <appender name="console" class="ch.qos.logback.core.ConsoleAppender">
```

```
    <encoder>
```

```
      <pattern>%d{HH:mm:ss.SSS} [%thread] %-5level %logger{36} - %msg%n</pattern>
```

```
    </encoder>
```

```
  </appender>
```

```
  <appender name="file" class="ch.qos.logback.core.FileAppender">
```

```
    <file>app.log</file>
```

```
    <encoder>
```

```
      <pattern>%d{HH:mm:ss.SSS} [%thread] %-5level %logger{36} - %msg%n</pattern>
```

```
    </encoder>
```

```
  </appender>
```

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```
<root level="debug">
  <appender-ref ref="console" />
  <appender-ref ref="file" />
</root>
</configuration>
```

3. Create a Java class that uses SLF4J for logging:

Write code for this.